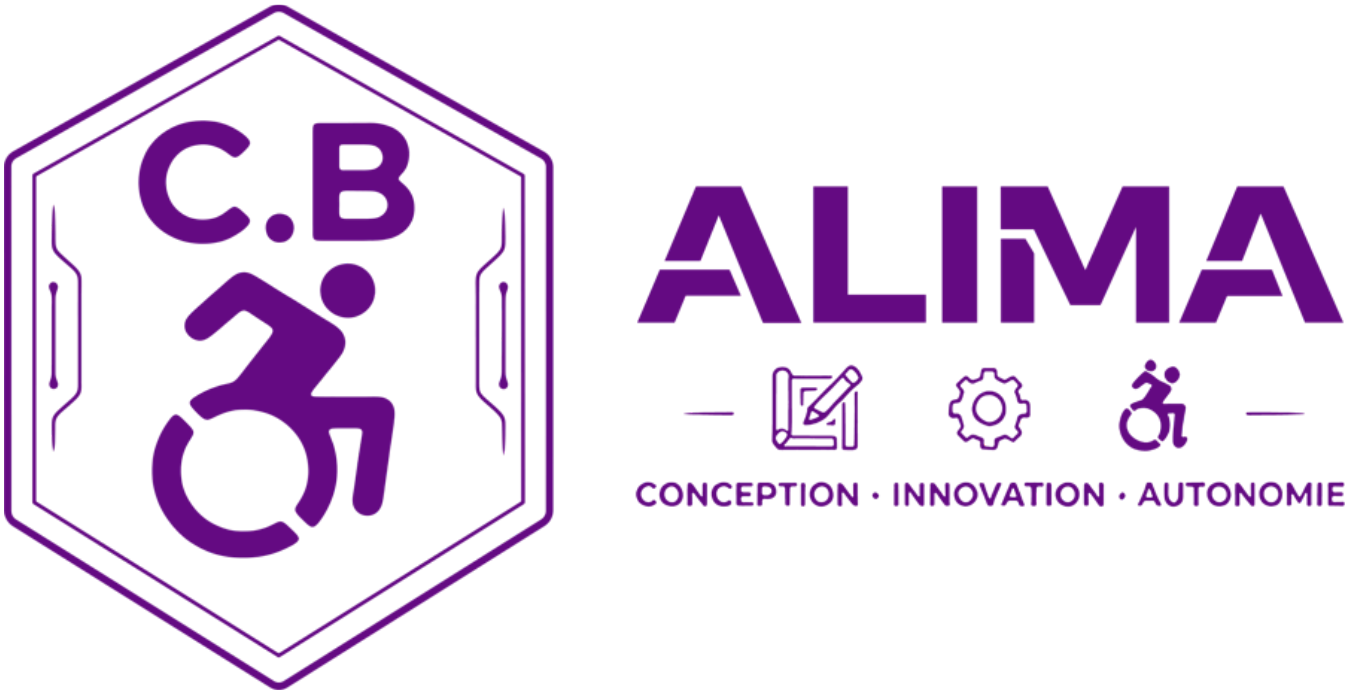
ALIMA Simply — Installer manual

CB ALIMA · Version October 2026, for Simply firmware 2.2 and later



CB ALIMA · Electronic design for adapted (PMR) vehicles · cb-alima.fr · contact@cb-alima.fr

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English translation provided for information. In case of any difference, only the French version of this manual is binding.

Overview

The ALIMA Simply is a button-operated auxiliary control board: it drives up to 6 vehicle functions (relays) from 4 switches, for drivers who cannot use the original controls.

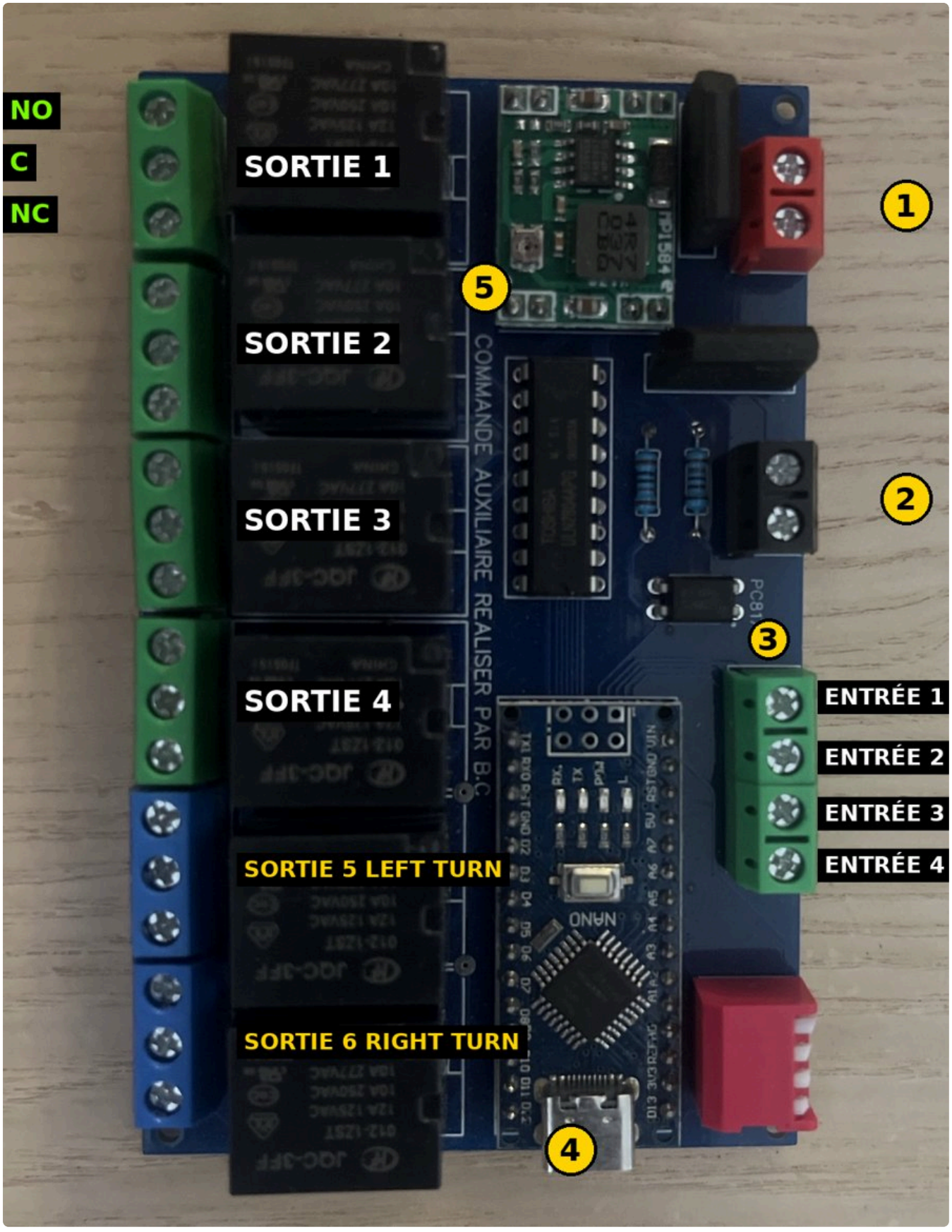
The names ENTRÉE (input) and SORTIE (output) are kept in French in this manual, exactly as they are printed on the plastic housing and shown in the ALIMA application.

Item	Specification
Microcontroller	Arduino Nano (ATmega328P)
Power supply	Vehicle 12 V, no polarity (diode bridge)

Item	Specification
Switch inputs	ENTRÉE 1 to ENTRÉE 4, connected to ground by the switch
Lights detection input	ENTRÉE 12 V (feux), no polarity, on the black terminal block
Outputs	SORTIE 1 to SORTIE 6 (relays). SORTIE 1 to 4: free (green terminal blocks). SORTIE 5 and 6: left and right turn signals, and hazard lights (blue terminal blocks)
Configuration link	USB-C (serial)

Board layout

Board seen from above, as in the photo (USB-C at the bottom). The names ENTRÉE 1…, SORTIE 1… and the markings C / NO / NC are those used in the ALIMA application: write them identically on the plastic housing. Terminal block colours are standardised.


NO
C
NC
SORTIE 1
SORTIE 2
SORTIE 3
SORTIE 4
SORTIE 5 LEFT TURN
SORTIE 6 RIGHT TURN
COMMANDE AUXILIAIRE REALISER PAR B.C
1
2
3
4
5
ENTRÉE 1
ENTRÉE 2
ENTRÉE 3
ENTRÉE 4

Mark	Terminal block	Name in the app	Role
1	Red, 2 screws	—	12 V power supply. Diode bridge: direction does not matter.
2	Black, 2 screws	ENTRÉE 12 V (feux)	Lights detection, on the two black pins, in parallel with a light (dipped beam, side lights, or number plate light if the front lights are multiplexed). No polarity.
3	Green, 4 screws	ENTRÉE 1 to ENTRÉE 4 (ENTRÉE 1 at the top in the photo)	Switches. Each input is connected to ground through the switch contact.
4	USB-C socket	—	Configuration with the ALIMA application.
5	Voltage regulator	—	Built-in, sealed potentiometer: do not touch.
—	6 terminal blocks, 3 screws (top to bottom: 4 green, then 2 blue)	SORTIE 1 to SORTIE 6 (SORTIE 1 at the top in the photo)	Relay outputs. Screw order, top to bottom in the photo: NO, C (common), NC. SORTIE 1 to 4 (green terminal blocks) are free. SORTIE 5 = left turn signal, SORTIE 6 = right turn signal (blue terminal blocks): these two functions are fixed.

Safety warnings

Installation must be carried out by a vehicle adaptation professional. A wiring error can cause a fire, the malfunction of a safety function or loss of control of the vehicle.

- Disconnect the negative battery terminal before any work, and wait for the circuits to discharge (airbags in particular).
- Never connect the board to a circuit related to active or passive safety (braking, steering, airbags, ABS/ESP) without ALIMA's written approval.

- Never directly control a light, a wiper or a regulated function without checking that the adaptation is allowed for this vehicle.
- The vehicle's original control must remain usable and have priority: the board must never prevent it from working.
- Each relay output must be protected by its own fuse, placed as close as possible to the power source.
- Do not exceed the maximum current of the relay contacts (10 A). For a higher load, use an external power relay.
- Never connect the USB cable to the computer while the vehicle is running or the board is driving loads: configure at standstill.
- Mount the board protected from water, heat and vibration, in the supplied housing, away from ignition cables and motors.
- Route cables in sleeves, with grommets where they pass through sheet metal, away from moving and hot parts.
- No modification of the board, its components or its firmware is accepted. The firmware is updated only through the ALIMA application, when it offers to do so.
- It is forbidden to touch the built-in, sealed potentiometer on the voltage regulator.
- Give the customer the user instructions and explain how the installation works and its limits.
- Connections must be soldered or crimped, then insulated with heat-shrink sleeving. No self-stripping connectors ("Scotchlok" type), no tapping into airbag harnesses (yellow sleeving) or vehicle communication lines (CAN, LIN, multiplexing).
- After installation, test every function with the vehicle stationary and the engine running, then during a road test carried out by the installer before handing over the vehicle.

Fuses and cable sizes

The fuse rating is chosen to protect the cable, not the load: it must be higher than the load current and lower than what the cable and the relay contact can withstand.

Circuit	Fuse	Cable size
Board power supply (red terminal block)	2 A	0.75 mm²
ENTRÉE 1 to ENTRÉE 4 (green terminal block)	No dedicated fuse	0.75 mm²

Circuit	Fuse	Cable size
ENTRÉE 12 V (feux) (black terminal block)	Fuse of the original lighting circuit	0.75 mm²
SORTIE 1 to SORTIE 6 (turn signals included)	Rated by the installer according to the load (table below), 10 A maximum	According to the load and the fuse (table below)

Each relay contact withstands 10 A maximum (value printed on the relay). Never fit a fuse above 10 A on an output, and never above 5 A if the cable stays at 0.75 mm².

Choosing the fuse for each output (12 V system). Current = power ÷ 12. The fuse must stay above the load current (at least 20 % margin) and never exceed what the cable can withstand.

Load on the output	Current	Minimum cable size	Recommended fuse
up to 24 W	2 A	0.75 mm²	3 A
up to 42 W	3.5 A	0.75 mm²	5 A
up to 72 W	6 A	1 mm²	7.5 A
up to 96 W	8 A	1 mm² (1.5 mm² recommended)	10 A
more than 96 W	more than 8 A	not allowed on the board	use an external power relay driven by the output

These values are indicative for short copper cables (less than 3 m). For a load with a high inrush current (motor, pump), use the starting current, or an external relay.

Indicative guide for automotive copper cable, to be confirmed according to the cable and the length of the run:

Cable size	Maximum fuse
0.75 mm²	5 A
1 mm²	7.5 A to 10 A
1.5 mm²	15 A
2.5 mm²	20 A
4 mm²	30 A

Fuse holder less than 30 cm from the source (battery or fuse box). Never replace a fuse with a higher rating.

Connecting the board

Battery disconnected during all wiring. Reconnect only once everything has been checked.

1. Fix the board in its plastic housing, protected from water and heat.
2. Power supply (mark 1, red terminal block): connect 12 V and ground with a 2 A fuse, in 0.75 mm². Direction does not matter.
3. ENTRÉE 12 V (feux) (mark 2, black terminal block): connect the two black pins in parallel with a light (dipped beam, side lights, or number plate light if the front lights are multiplexed). Direction does not matter.
4. ENTRÉE 1 to ENTRÉE 4 (mark 3, green terminal block): each screw is one input. The switch contact connects the screw to ground.
5. SORTIE 5 (left turn signal) and SORTIE 6 (right turn signal), blue terminal blocks: connect to C and NO (or NC depending on the installation), with their fuse. These two outputs are reserved for the turn signals and hazard lights.
6. SORTIE 1 to SORTIE 4, green terminal blocks: connect each load to C and NO (or NC as required), with its fuse. Screw order: NO, C, NC.
7. Reconnect the battery, then follow the commissioning procedure with the ALIMA application (next section) to test each output.

Terminal block	Colour	Screws	Connection
Power supply	Red	2	12 V and ground, 2 A fuse, no polarity
ENTRÉE 12 V (feux)	Black	2	In parallel with a light, no polarity
ENTRÉE 1 to ENTRÉE 4	Green	4	To ground through the switch

Terminal block	Colour	Screws	Connection
SORTIE 1 to SORTIE 4 (free)	Green	3 each	NO / C / NC
SORTIE 5 (left turn signal)	Blue	3	NO / C / NC
SORTIE 6 (right turn signal)	Blue	3	NO / C / NC

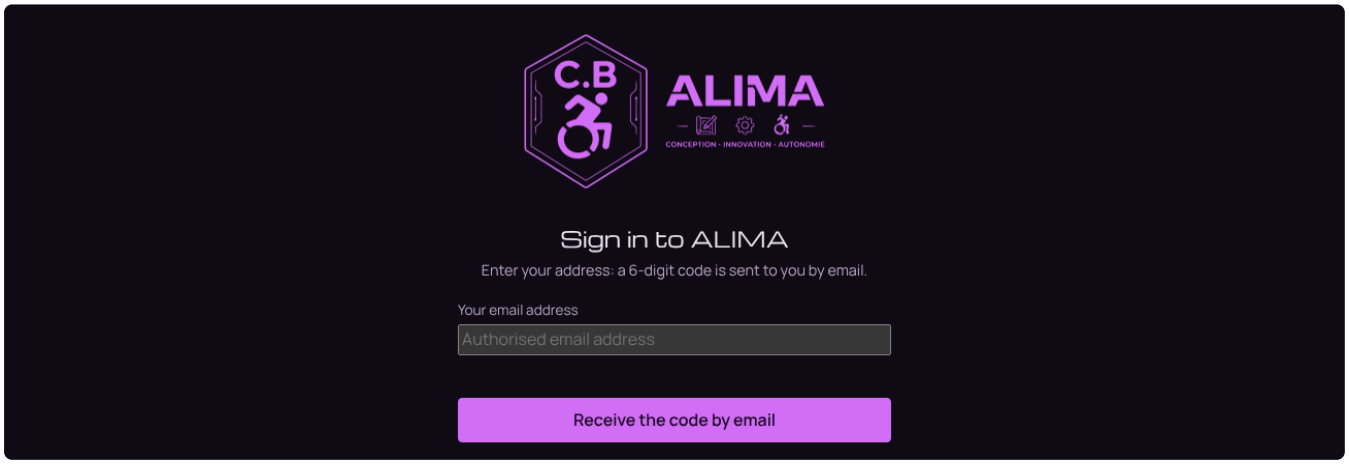
Configuration

The board is configured through the USB-C socket (mark 4) with the ALIMA application, vehicle stationary. The application uses the same names as the housing: ENTRÉE 1 to 4, ENTRÉE 12 V (feux), SORTIE 1 to 6. These names are never translated, whatever language is chosen.

Getting the application and signing in

The ALIMA application (Windows or Mac) is downloaded from the installer area of the cb-alima.fr website. Access is restricted to installers whose e-mail address has been authorised by CB ALIMA or its distributor.

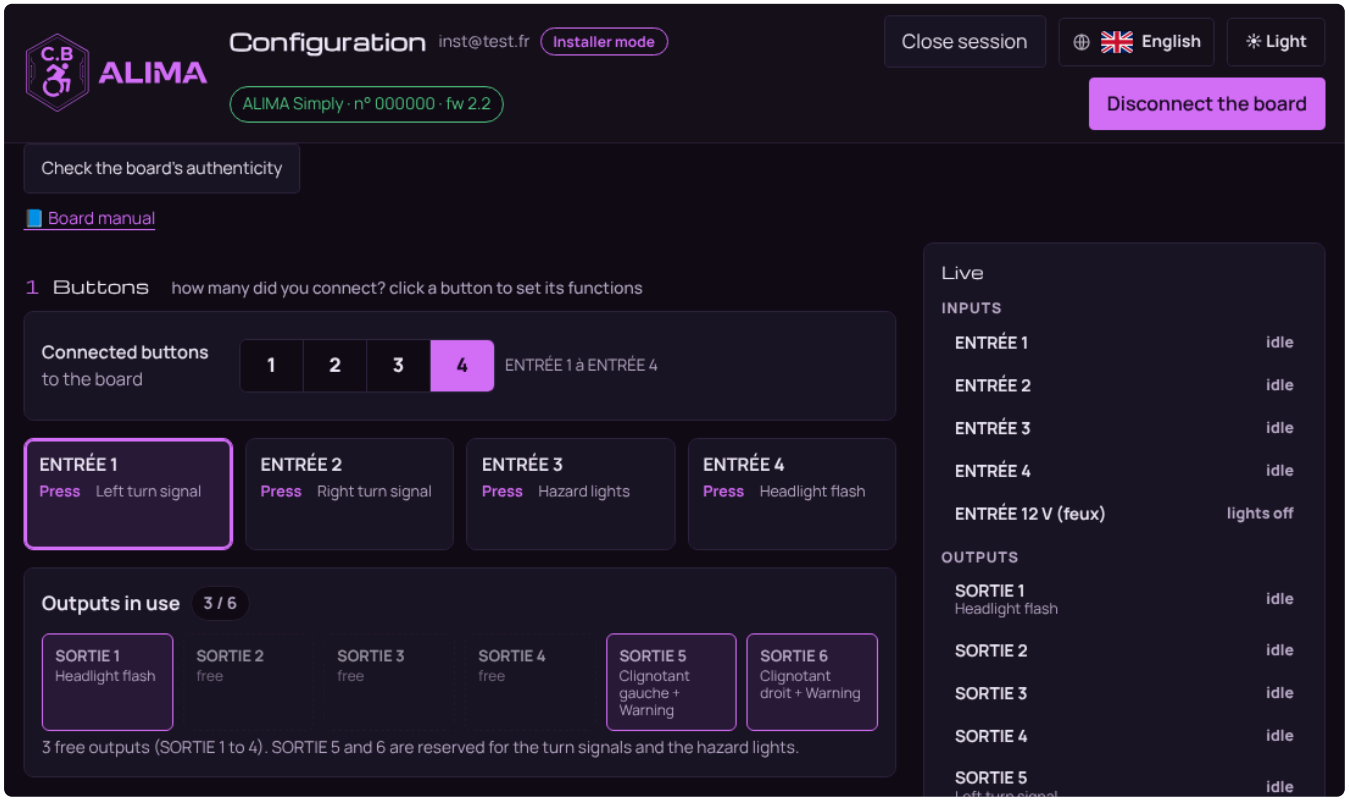
At start-up, enter your e-mail address, press “Receive the code by email” and type in the 6-digit code you receive. Without an internet connection, the application can still be used in installer mode for 15 days after the last successful sign-in on that computer. The application updates itself: accept the update when it is offered. The language (8 languages) and the light or dark theme are chosen at the top right.



Commissioning

Step 1. Connect the board. Power the board, then connect it to the computer with a USB-C data cable (not a charge-only cable). Click “Connect a board” and choose the port. Opening the port restarts the board: the outputs drop out. Never connect while driving.

The green badge shows the model (ALIMA Simply), the serial number and the firmware version. On the right, the “Live” panel shows the state of each ENTRÉE, of ENTRÉE 12 V (feux) and of each SORTIE.



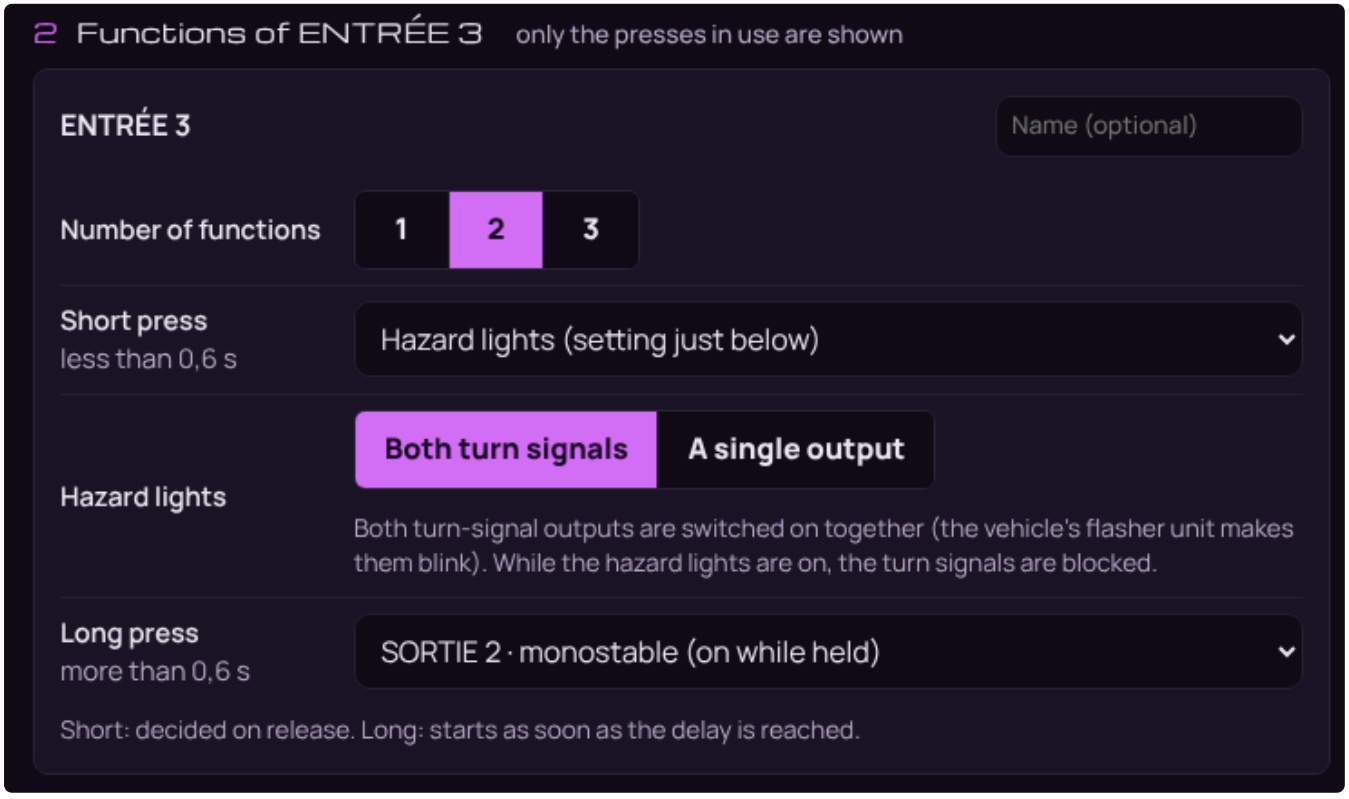
Factory configuration: ENTRÉE 1 = left turn signal (SORTIE 5), ENTRÉE 2 = right turn signal (SORTIE 6), ENTRÉE 3 = hazard lights, ENTRÉE 4 = headlight flash (SORTIE 1).

Step 2. Check the board. With internet access, the application checks the board against the ALIMA register:

- a banner shows the warranty status: number of days left to send the installation record, or “under warranty until …” once the warranty is activated;
- a red banner saying the board is not in the ALIMA register means the board is not recognised: do not install it and contact CB ALIMA;
- the “Check the board's authenticity” button confirms that the board is a genuine ALIMA board;
- if the application shows “Update the firmware”, run the update before setting up the board. The serial number is kept. A board moving from an earlier version to version 2.2 returns to the factory configuration: set it up again, and check that the turn signals are connected to SORTIE 5 and SORTIE 6.

Step 3. Buttons (section 1). Choose the number of connected buttons (1 to 4): they match ENTRÉE 1 to ENTRÉE n. Click the tile of an ENTRÉE to set its functions. The “Outputs in use” panel shows how many of the 6 SORTIES are taken; SORTIE 5 and 6 are reserved for the turn signals and hazard lights.

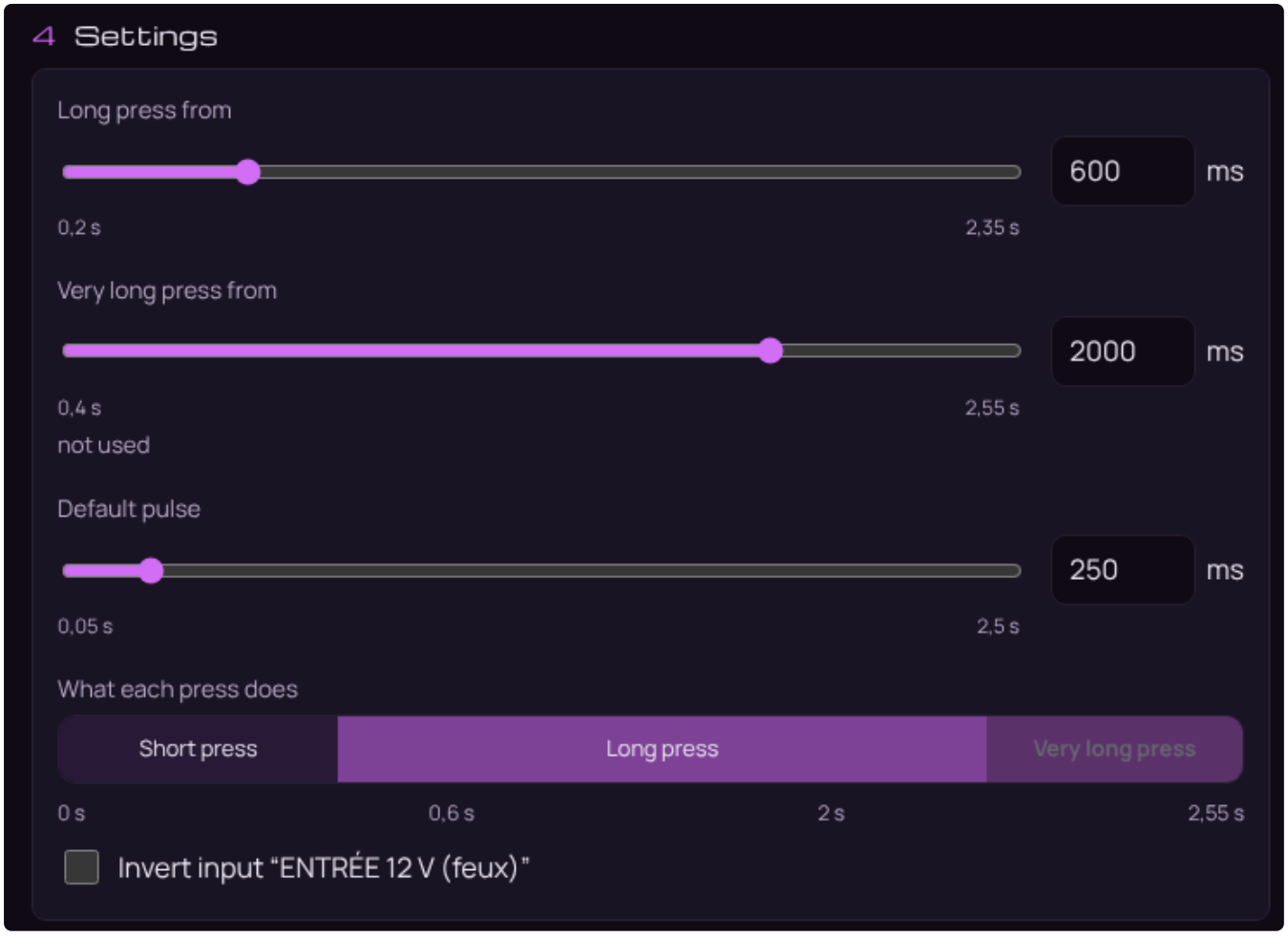
Step 4. Functions of the ENTRÉE (section 2). Choose the number of functions of the button (1 to 3: short, long, very long press), then the function of each press from the list. Each function shows the SORTIE it controls.



- **Left / right turn signal**: SORTIE 5 / SORTIE 6. The output stays on; the vehicle's flasher unit makes the lights blink.
- **Hazard lights**: either both turn signal outputs on together, or a single output. While the hazard lights are on, the turn signals are locked.
- **Headlight flash / main beam**: lights off = headlight flash, lights on = main beam (lights read on ENTRÉE 12 V (feux)). Also available as "headlight flash only" and "main beam only". SORTIE 1 by default.
- **SORTIE 1 to 4**: latching (toggles at each press), momentary (on while the button is held) or pulse (horn tap…).
- **Cut everything**: releases all outputs.

Step 5. Outputs (section 3). The "Board outputs" block is folded: it summarises what each SORTIE does. Open it to name the outputs (optional, kept on the computer and not in the board) and to choose the main beam output. It opens by itself if it contains an error.

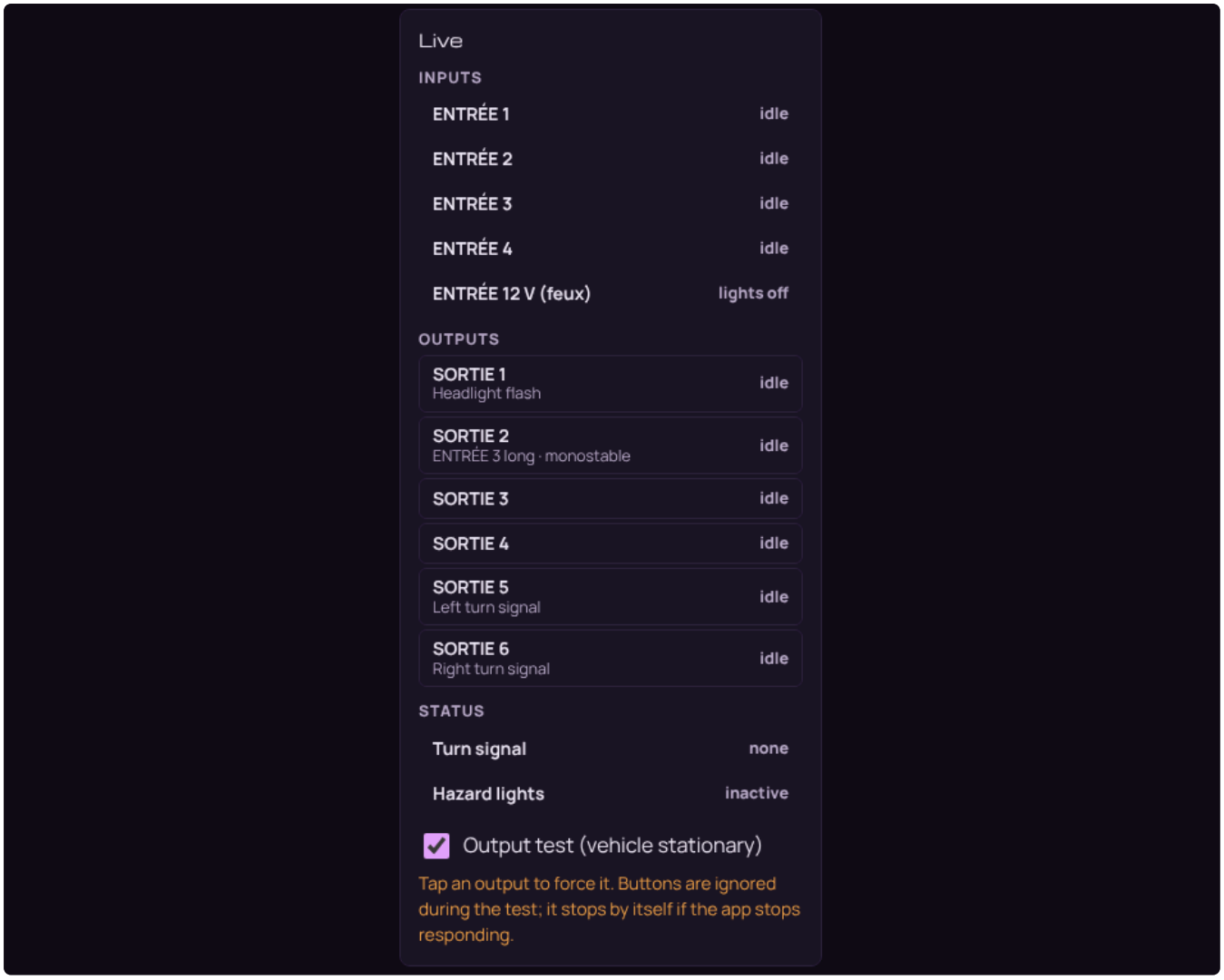
Step 6. Timing settings (section 4). Each duration is set with the slider or by typing the exact value: long press from 600 ms (minimum 200 ms), very long press from 2000 ms (always at least 200 ms more than the long press), default pulse 250 ms (50 to 2500 ms). The "What each press does" strip shows live which function is triggered depending on how long the button is pressed. A box inverts ENTRÉE 12 V (feux).



Step 7. Send to the board. Press “Send to the board”. The badge changes from “Modified, not sent yet” to “Same as the board”. If a red “To fix before sending” frame appears, correct the point shown: sending is blocked while an error remains.



Step 8. Test the outputs, vehicle stationary. Tick “Output test (vehicle stationary)”, then tap a SORTIE to force it on and check that the right load responds. The buttons are ignored during the test, which stops by itself if the application stops responding.



Step 9. Check and save. Untick the test, try each ENTRÉE with its switch, then click “Export” to save the configuration to a file: this file goes with the installation record. Disconnect the board, then unplug the USB cable.

“Import” copies a saved configuration to another board, “Read the board” reads back what is stored in the board, “Factory settings” restores the original configuration. The “Board manual” link opens this manual. “Close session” signs the account out of the computer.

Checks before handing over the vehicle

- [ ] Wiring visually checked: no pinched or stripped cable, no contact with hot or moving parts.
- [ ] All fuses fitted with the correct rating.
- [ ] Each ENTRÉE (1 to 4) triggers the correct function with its switch.
- [ ] Each SORTIE controls the correct load, and nothing else (application output test).
- [ ] The vehicle's original control still works.

- [ ] ENTRÉE 12 V (feux): the function responds correctly with the lights on, then off.
- [ ] No warning light on the dashboard, no abnormal battery drain when parked.
- [ ] Manual given to the customer and operation explained.

Symptom	Check
Board does not respond	Power fuse, ground, +12 V supply
A relay does not engage	Output fuse, load wiring
Function triggers by itself	Interference: move cables away, check the ground

Installation record

To be filled in and signed by the installer for each vehicle, then kept with the configuration file exported from the ALIMA application. It covers the identification of the vehicle and the board (VIN, serial number, firmware version), the function and fuse of each SORTIE, the checks before handover, the road test and the information given to the customer.

The record is the **last page of this manual**, alone on its page so it can be printed separately. Its fields can be filled in directly in a PDF reader (Adobe Acrobat Reader, web browser). It can also be downloaded on its own: ALIMA Simply installation record (fillable PDF).

Send it to contact@cb-alima.fr within 10 days of installation, together with the exported configuration file: this is what activates the 2-year warranty (see “2-year warranty”).

Liability and limits of use

Installation must be carried out by a qualified professional. By installing the board, the installer acknowledges having read this manual and accepts its conditions.

- The installer is solely responsible for the installation: choice and fitting of fuses, cable sizes, quality of connections, tests and handover of the vehicle to the customer.
- The installer is responsible for compliance with the regulations applicable to the modified vehicle (inspection, declaration or type approval of the adaptation, informing the customer's insurer, where applicable).
- The ALIMA Simply board is not a safety device. It must never be connected to a safety circuit (braking, steering, airbag, traction). Tests are carried out with the vehicle stationary.

- Any modification of the board, its components or its firmware is prohibited, including adjusting the sealed potentiometer of the voltage regulator. It voids the warranty and releases CB ALIMA from all liability.
- CB ALIMA cannot be held liable for damage resulting from an installation that does not comply with this manual, misuse, overloading of the outputs or modification of the board.
- Within the limits permitted by law, CB ALIMA is not liable for indirect damage (vehicle downtime, loss of business, removal and refitting costs, etc.). Its liability is limited, at its option, to the repair, replacement or refund of the board.
- The ALIMA Simply board is intended exclusively for installation professionals. It is not sold to private individuals for self-installation.
- The board configuration (choice of functions, assignment of ENTRÉES and SORTIES, timings) is carried out and validated by the installer, who is responsible for it. The ALIMA application is provided as is, as a configuration tool.
- The installer fills in the installation record, keeps it with the exported configuration file, and provides it to CB ALIMA in the event of a claim. Without an installation record, the conformity of the installation cannot be verified.
- Any incident, defect or abnormal behaviour must be reported to CB ALIMA without delay, and use of the function concerned suspended until it is resolved.
- These limitations do not apply in the event of personal injury, gross negligence or wilful misconduct, or where the law prohibits them.

2-year warranty

The ALIMA Simply board is guaranteed against manufacturing and component defects: 8 months from its shipment by CB ALIMA, then, once the warranty is activated, 2 years from the activation date.

Activation. The 2-year warranty is activated by CB ALIMA on receipt of the installation record, complete and signed (with the end customer's name and the vehicle), sent to contact@cb-alima.fr within 10 days of installation, and at the latest before the end of the 8 months following shipment. The date the e-mail is received is authoritative; CB ALIMA acknowledges receipt. The ALIMA application shows the warranty status of the connected board. A record sent late, incomplete or after a failure has occurred does not allow activation; without activation, the warranty ends 8 months after shipment. Statutory warranties remain applicable.

What the warranty covers. Repair or replacement of the defective board, at CB ALIMA's option. A repaired or replaced board keeps the remaining warranty period.

How to claim. Write to contact@cb-alima.fr with the board serial number, the invoice and a description of the problem (photos of the installation if possible). The return of the board is agreed with CB ALIMA. Shipping of the board to CB ALIMA is paid by the customer; CB ALIMA pays for returning the repaired or replaced board if the warranty applies.

Exclusions. The warranty does not cover:

- an installation that does not comply with this manual: wrong connection, polarity or voltage outside specification, missing or unsuitable fuse;
- overloading of the outputs (more than 10 A per relay contact) or a short circuit;
- water, moisture, impacts, excessive heat, corrosion or surges on the vehicle's electrical system;
- any modification or repair by a third party, including adjusting the sealed regulator potentiometer;
- firmware not supplied by CB ALIMA;
- an accident, fire, transport damage or force majeure;
- normal wear of relays, terminals, connectors and cables;
- a board whose serial number is missing or illegible;
- removal and refitting costs, labour and vehicle downtime.

This commercial warranty is in addition to, and does not exclude, the statutory warranties.

Governing law. These conditions are governed by French law. Between professionals, any dispute falls within the jurisdiction of the competent courts for the registered office of CB ALIMA, after an attempt at amicable settlement.

This manual is an integral part of the product. It must be read before installation, kept by the installer and handed over with the vehicle. Only the current version, available from CB ALIMA, is binding; in case of any difference between this translation and the French version, the French version prevails. ALIMA Simply installer manual, version 2026-10, firmware 2.2.

CB ALIMA · Clément BILQUE, sole trader (entrepreneur individuel, EI) · Tel.: +33 6 95 68 69 56 · contact@cb-alima.fr · cb-alima.fr

Installation record — ALIMA Simply

To be filled in and signed by the installer for each vehicle, then kept with the configuration file.
Send to contact@cb-alima.fr within 10 days of installation: condition of the 2-year warranty.

1. Identification

Installing company

Technician

Installation date

End customer (first and last name)

Vehicle make and model

Registration number

VIN

Board serial number

Firmware version

Name of the exported configuration file

2. Connections and fuses

Circuit	Connected function	Load (W)	Fuse (A)	Cable (mm²)
Power supply	Board power supply		2	
SORTIE 1				
SORTIE 2				
SORTIE 3				
SORTIE 4				
SORTIE 5	Left turn signal			
SORTIE 6	Right turn signal			

3. Checks before handing over the vehicle

- ☐ Wiring checked: no pinched or stripped cable, no contact with hot or moving parts
- ☐ Connections soldered or crimped and insulated, no self-stripping connectors
- ☐ All fuses fitted with the correct rating, as close as possible to the source
- ☐ Each ENTRÉE (1 to 4) triggers the correct function with its switch
- ☐ Each SORTIE controls the correct load and nothing else (application output test)
- ☐ ENTRÉE 12 V (feux): the function responds correctly with lights on, then off
- ☐ The vehicle's original control still works and keeps priority
- ☐ No warning light on the dashboard, no abnormal battery drain when parked
- ☐ Tests stationary with engine running, then road test, carried out by the installer
- ☐ Configuration exported and kept with this record
- ☐ Operation and limits explained to the customer, manual handed over

4. Remarks

5. Validation

Name and position of signatory

Date

Signature and company stamp